

Question Number	Answer	Additional guidance	Mark
4(a)(i)	6.4 (mm ³) (1)	= 1.3r x 3.14 x 1.15 ³ = 6.4	(1)

Question Number	Answer	Additional guidance	Mark
4(a)(ii)	An answer that includes two of the following points: - contains {more cytoplasm (than sperm cell) / contains named organelles} (1) - contains {oil / lipid / triglyceride} (droplets) (1) (which) provides {energy / ATP / monomers} (for developing embryo) (1)	e.g. endoplasmic reticulum, Golgi apparatus, more mitochondria, ribosomes accept contain cortical granules ignore nucleus / zona pellucida ignore mitochondria unqualified ignore food / nutrients ACCEPT (to supply materials) for {mitosis / cell division / development of embryo / growth of embryo} ignore {make/create/produce} energy accept reasons as to why the egg cell has {ER / ribosomes / Golgi apparatus} linked to protein synthesis	(2)

Question Number	Answer	Mark
4(b)(i)	The only correct answer is B metaphase <i>A is not correct because the cell is in metaphase</i> <i>C is not correct because the cell is in metaphase</i> <i>D is not correct because the cell is in metaphase</i>	(1)

Question Number	Answer	Additional guidance	Mark
4(b)(ii)	An answer that includes the following point: a {stain / dye} was applied (to the cell) to make the {DNA / chromatin / chromosomes / spindle fibres / histone proteins} visible (1)	ignore stain unqualified accept {stain / dye} linked to a named structure inside the cell seen during mitosis e.g. add stain to chromosomes	(1)

Question Number	Answer	Additional guidance	Mark
4(b)(iii)	An answer that includes three of the following points: Similarities • both contain diploid cells (1) • both contain cells dividing by mitosis (1) Differences • morula contains totipotent cells whereas blastocyst contains pluripotent cells (1) • morula {is a solid ball of / mass of} cells whereas blastocyst {is a hollow ball of / contains many} cells (1)	Full marks can only be awarded if there is a similarity in the answer Can piece together from adjacent sentences accept both have diploid number of chromosomes ignore they have both formed by mitosis accept morula is totipotent whereas blastocyst is pluripotent accept morula has {undifferentiated cells / all genes switched on} whereas blastocyst {doesn't / has some differentiated cells / genes switched off} accept labelled diagram showing these aspects accept blastocyst contains {more cells / is larger} (than morula) / converse accept blastocyst contains a {trophoblast / inner cell mass / fluid filled cavity} whereas morula doesn't	(3)

Question Number	Answer	Additional guidance	Mark
4(c)(i)	A description that includes two of the following points: • flagellum rotates (1) • {energy/ATP} released by the mitochondria (in the mid piece) (1) • streamlined shape to reduce {resistance / friction} (1)	accept has a flagellum unqualified ignore has a tail unqualified accept tail {has whip like motion / rotates} do not accept {makes / generates / produces} energy accept provides energy accept {makes / generates / provides / produces} ATP	(2)

Question Number	Answer	Additional guidance	Mark
4(c)(ii)	An answer that includes the following points: • as time (after release into water) increases, the sperm {velocity / speed} decreases (1) • sperm have a greater {velocity / speed} in March (1)	Accept negative correlation ignore sperm cells slow down after being added to water accept converse for February ignore references to numbers of sperm	(2)